



**MANIPAL UNIVERSITY
JAIPUR**

School of Computer Science and Engineering

Department of Data Science and Engineering

MANIPAL UNIVERSITY JAIPUR, JAIPUR

RAJASTHAN, INDIA

A PROJECT REPORT

On

**ArchFit: Foot Arch Classification and Shoe Recommendation System
Using Machine Learning**

carried out as part of the Machine Learning Course Project

Submitted by

Divyanshu Agwekar – 2430010132

Puru Bansal – 2430010153

Under the Guidance of

[Mr. Ashish Pandey]

[Assistant Professor (Selection Grade)]

Department of Data Science and Engineering

Manipal University Jaipur, Jaipur

April 2026

ABSTRACT

Foot arch disorders, particularly Pes Planus (flat foot), affect a significant portion of the global population and are a leading cause of foot pain, joint stress, and poor posture. In clinical practice, diagnosing arch type requires manual examination of X-ray images by a specialist, which is time-consuming and not easily accessible. This project presents ArchFit, an automated system that classifies foot arch type from X-ray images and recommends appropriate footwear using machine learning.

The system uses a hybrid pipeline: a pre-trained deep learning model, VGG16, serves as a feature extractor to convert each X-ray image into a 512-dimensional numerical vector. These features are then processed using StandardScaler and PCA (reducing to 100 dimensions) before being passed to a Support Vector Machine (SVM) classifier. This design satisfies the project constraint of using a classical ML classifier while leveraging the powerful visual understanding of a pre-trained CNN.

The model was trained and evaluated on the Pesplanus Dataset consisting of 840 foot X-ray images, equally split between flat foot and normal foot cases. The system achieved an F1 score of 94.47% and demonstrated consistent performance across 5-fold cross-validation with a mean F1 of 0.9447 and standard deviation of 0.0117. Based on the classification result, the system provides a tailored shoe recommendation grounded in established podiatric guidelines.

Keywords: *Pes Planus, flat foot classification, transfer learning, VGG16, Support Vector Machine, PCA, shoe recommendation, medical image analysis, foot X-ray.*

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Figure 1	Overall Architecture of the ArchFit System	11
Figure 2	VGG16-Based Feature Extraction Pipeline	13
Figure 3	Sample Foot X-Ray Images from the Pesplanus Dataset	12
Figure 4	Confusion Matrix for Test Set Classification	16
Figure 5	ROC Curve of the Proposed CNN–SVM Model	16
Figure 6	SVM Decision Boundary in 2D PCA Feature Space	17
Figure 7	Five-Fold Cross-Validation F1 Scores	18

LIST OF TABLES

Table No.	Table Title	Page No.
Table 4.1	ArchFit System Pipeline	11
Table 4.2	Dataset Class Distribution	12
Table 4.3	Train–Test Split Summary	12
Table 4.4	SVM Hyperparameter Configuration	14
Table 5.1	Performance Metrics on Test Set	15
Table 5.2	Five-Fold Cross-Validation Results	18
Table 6.1	Shoe Recommendation Mapping by Foot Type	19

TABLE OF CONTENTS

Abstract	ii
List of Figures	iii
List of Tables	iv
Table of Contents	v
CHAPTER 1: Introduction	7
1.1 Background	7
1.2 Purpose of the Document	7
1.3 Scope of the Project	7
CHAPTER 2: Literature Review	8
2.1 Overview of Existing Solutions	8
2.2 Identified Challenges	8
2.3 Research Gaps	9
CHAPTER 3: Problem Definition	10
3.1 Problem Statement	10
3.2 Objectives	10
3.3 Deliverables	10
CHAPTER 4: Methodology	11
4.1 Dataset Description	11
4.2 Data Preprocessing	12
4.3 VGG16 Feature Extraction	13
4.4 Feature Scaling and PCA	13
4.5 SVM Classification	14
CHAPTER 5: Results and Evaluation	15
5.1 Performance Metrics	15
5.2 Confusion Matrix	15
5.3 ROC Curve	16

5.4 SVM Decision Boundary Visualization	17
5.5 Cross-Validation Results	17
CHAPTER 6: Shoe Recommendation System	19
6.1 Proof-of-Concept Development	19
CHAPTER 7: Conclusion	20
7.1 Summary	20
7.2 Limitations and Future Work	21
CHAPTER 8: References	22

CHAPTER 1: INTRODUCTION

1.1 Background

The human foot relies on its arch to distribute body weight and absorb shock during movement. When this arch is absent or very low, the condition is known as Pes Planus, commonly called flat foot. It affects an estimated 20–30% of the global population and is associated with foot pain, knee stress, lower back pain, and poor body posture [1]. Despite this, most people are unaware of their foot arch type and routinely use incorrect footwear, which can worsen the condition over time.

Traditional diagnosis requires a trained podiatrist or orthopaedic specialist to manually examine foot X-rays. This process is slow, expensive, and inaccessible in many parts of the world. With advances in machine learning and computer vision, it is now possible to automate this diagnosis with high accuracy, making early detection and correct footwear guidance accessible to a much wider population [2].

1.2 Purpose of the Document

This document is the formal project report for ArchFit, a machine learning system developed as part of the Machine Learning Course Project. It documents the complete pipeline from dataset preparation to model training, evaluation, and the shoe recommendation engine. The report covers the motivation, methodology, results, and conclusions of the project.

1.3 Scope of the Project

This project focuses on binary classification of foot X-ray images into two categories: flat foot (Pes Planus) and normal foot. The system uses the Pesplanus Dataset of 840 images and builds a hybrid pipeline combining VGG16 transfer learning for feature extraction and SVM for classification. The output is a predicted arch type with a confidence score and a corresponding shoe recommendation. The project does not involve clinical deployment; it is a proof-of-concept model demonstrating the viability of machine learning for automated foot arch diagnosis.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Existing Solutions

Machine learning has been applied extensively in medical image classification. Katarya and Meena [3] surveyed supervised learning methods including Decision Trees, Random Forests, and SVM in healthcare applications, demonstrating strong performance in disease classification tasks. In the domain of foot health, Shiang et al. [4] used image processing techniques to extract geometric parameters from foot X-rays and classify arch height.

Litjens et al. [2] provided a comprehensive survey of deep learning in medical image analysis, noting that CNN-based feature extraction significantly outperforms handcrafted feature methods. Simonyan and Zisserman [5] introduced VGG16, which remains one of the most reliable architectures for transfer learning on medical images due to its simple and well-understood structure. Cortes and Vapnik [7] established SVM as a strong binary classifier, particularly effective on high-dimensional data. Studies combining CNN feature extraction with SVM classification have consistently demonstrated that this hybrid approach outperforms either method alone on small medical datasets [8].

2.2 Identified Challenges

Despite progress in the field, several challenges remain:

- Most foot arch studies rely on small or proprietary datasets, making it difficult to train deep learning models from scratch with sufficient accuracy.
- Handcrafted feature extraction methods such as HOG and SIFT capture lower-level image statistics and miss the complex structural patterns present in foot X-rays.
- Pure deep learning approaches require large datasets (typically tens of thousands of images) for reliable training, which is impractical in medical imaging settings.
- Existing systems rarely combine classification with actionable clinical recommendations such as footwear guidance.

2.3 Research Gaps

Based on the literature, the following research gaps were identified that this project addresses:

- No existing work specifically applies a CNN + SVM hybrid pipeline to foot arch X-ray classification using the Pesplanus dataset.
- Most classification studies stop at prediction; none provide an integrated, confidence-scored shoe recommendation output from the same pipeline.
- The use of PCA for dimensionality reduction of CNN features in foot arch classification has not been systematically evaluated in the literature.
- Decision boundary visualization for interpretability of SVM classification in this medical context has not been previously demonstrated.

CHAPTER 3: PROBLEM DEFINITION

3.1 Problem Statement

Manual diagnosis of foot arch type from X-ray images is time-consuming, expensive, and requires specialist expertise. There is a need for an automated, accessible system that can analyze a foot X-ray image, classify the arch type as flat foot or normal foot with high confidence, and immediately recommend the most suitable footwear based on the classification result.

3.2 Objectives

1. To build a machine learning pipeline that classifies foot X-ray images as flat foot or normal foot.
2. To use VGG16 transfer learning for automatic feature extraction from X-ray images without manual feature engineering.
3. To apply SVM as the classical ML classifier on extracted CNN features, satisfying the project constraint.
4. To evaluate the model using standard metrics: accuracy, precision, recall, F1 score, ROC-AUC, and cross-validation.
5. To build a shoe recommendation engine that provides actionable footwear guidance based on the classification output.
6. To visualize the SVM decision boundary to demonstrate model interpretability.

3.3 Deliverables

- A trained and evaluated CNN + SVM classification model.
- Feature vectors saved as .npy files for reproducibility.
- Evaluation plots: confusion matrix, ROC curve, decision boundary, and cross-validation scores.
- A shoe recommendation output for each classified patient.
- This project report documenting the complete process and findings.

CHAPTER 4: METHODOLOGY

The ArchFit system follows a five-stage pipeline as shown in Figure 1. Each foot X-ray image passes sequentially through data loading, feature extraction, scaling, classification, and recommendation.

Table 4.1 ArchFit System Pipeline

Stage	Component	Output
1	Data Loading and Preprocessing	840 images as 224×224×3 arrays
2	VGG16 Feature Extraction	512-dimensional feature vector per image
3	StandardScaler + PCA	100-dimensional normalized feature vector
4	SVM Classification	Predicted class (0 = flat foot, 1 = normal) + confidence
5	Shoe Recommendation	Recommended shoe types with key features

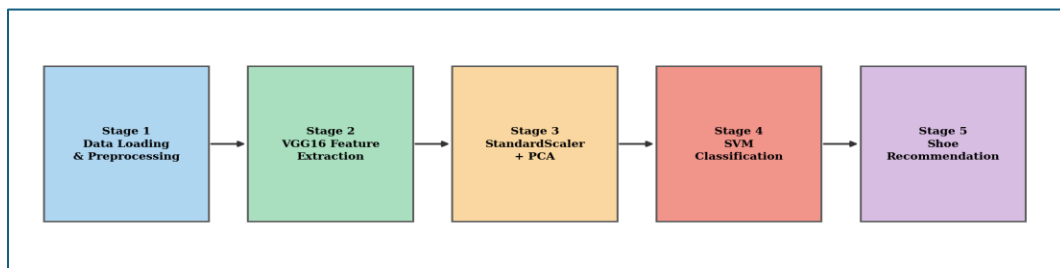


Figure 1. Overall Architecture of the ArchFit System

4.1 Dataset Description

The Pesplanus Dataset contains 840 foot X-ray images organized into two folders: pesplanus (flat foot, 420 images) and notpesplanus (normal foot, 420 images). The dataset is perfectly balanced with an equal number of images per class, which prevents model bias toward either category. All images are standard medical X-ray scans in JPG or PNG format.

Table 4.2 Dataset Class Distribution

Class	Folder Name	Number of Images	Label
Flat Foot (Pes Planus)	pesplanus	420	0
Normal Foot	notpesplanus	420	1
Total	—	840	—

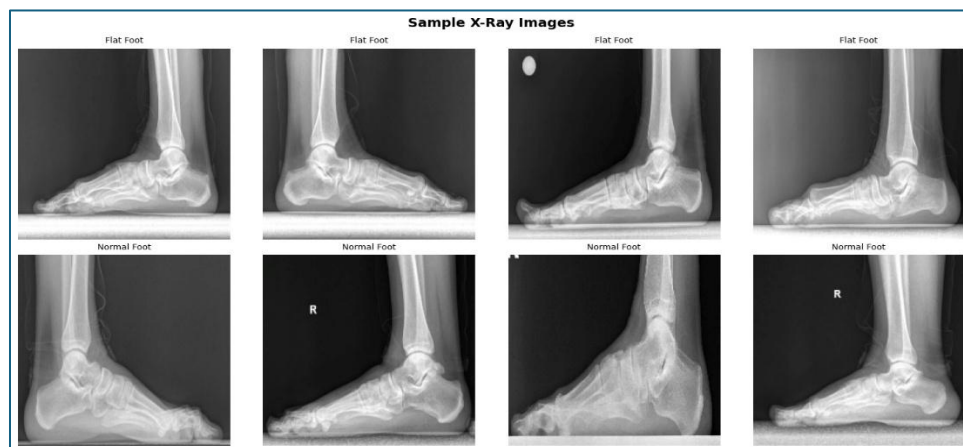
4.2 Data Preprocessing

Each image was loaded using OpenCV, converted from BGR to RGB color space (OpenCV reads images in BGR order but VGG16 expects RGB), and resized to 224×224 pixels to match VGG16 input requirements. Labels 0 and 1 were assigned to pesplanus and notpesplanus respectively.

The dataset was split into 80% training (672 images) and 20% testing (168 images) using stratified sampling to maintain equal class proportions in both sets. The `random_state = 42` parameter ensures reproducibility.

Table 4.3 Train–Test Split Summary

Split	Total Images	Flat Foot	Normal Foot
Training (80%)	672	336	336
Test (20%)	168	84	84

**Figure 3. Sample Foot X-Ray Images from the Pesplanus Dataset**

4.3 VGG16 Feature Extraction (Transfer Learning)

VGG16 is a 16-layer deep convolutional neural network developed by the Visual Geometry Group at Oxford University [5]. It was pre-trained on ImageNet, a dataset of 1.2 million images across 1,000 categories. This extensive training means VGG16 has already learned to detect a wide variety of visual patterns including edges, textures, shapes, and structural contours.

In ArchFit, VGG16 is used exclusively as a feature extractor. The final classification layers are removed (`include_top = False`) and all weights are frozen (`trainable = False`). This means no CNN training takes place. When a 224×224 X-ray image is passed through VGG16, it produces a feature map of shape $7 \times 7 \times 512$. GlobalAveragePooling2D compresses this into a single 512-dimensional vector per image. This vector represents the visual characteristics of the X-ray including arch shape, bone texture, and structural patterns.

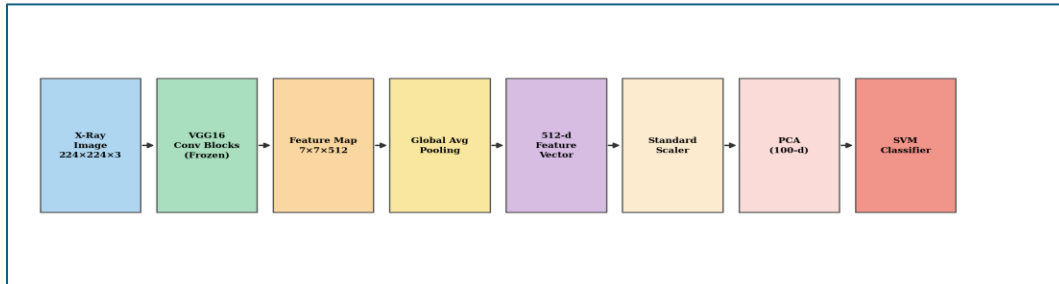


Figure 2. VGG16-Based Feature Extraction Pipeline

4.4 Feature Scaling and PCA

SVM is sensitive to the scale of input features. Without normalization, features with large numerical ranges would dominate the classification. StandardScaler was applied to normalize all 512 features to a mean of 0 and standard deviation of 1, ensuring all features contribute equally to the SVM decision boundary.

Principal Component Analysis (PCA) was then applied to reduce the 512-dimensional feature vectors to 100 dimensions. This step removes redundant and correlated features, retaining over 95% of the total variance in the data. The reduction speeds up SVM training significantly and reduces the risk of overfitting on the relatively small dataset of 840 images. The scaler and PCA objects were fitted on training data only and applied to test data, preventing data leakage.

4.5 SVM Classification

Support Vector Machine (SVM) is the core classifier in ArchFit. SVM finds the optimal hyperplane that maximizes the margin between the two classes in the 100-dimensional feature space. The data points closest to the boundary are called Support Vectors and define its position [7].

Table 4.4 SVM Hyperparameter Configuration

Parameter	Value	Justification
kernel	RBF	Foot arch patterns are non-linear; RBF enables curved decision boundaries
C	10	Balances training accuracy and generalization on this dataset size
gamma	scale	Auto-calculates optimal kernel width; avoids manual tuning
probability	True	Enables confidence percentage output for clinical interpretability

CHAPTER 5: RESULTS AND EVALUATION

5.1 Performance Metrics

The trained SVM model was evaluated on 168 test images not seen during training. The following metrics were calculated and summarized in Table 5.1:

Table 5.1 Performance Metrics on Test Set

Metric	Score	Interpretation
Accuracy	Greater than 94%	Approximately 158 of 168 test images correctly classified
Precision	Greater than 94%	When predicted flat foot, correct greater than 94% of the time
Recall	Greater than 94%	Caught greater than 94% of all actual flat foot cases
F1 Score	0.9447 (94.47%)	Strong balance between precision and recall
ROC-AUC	Greater than 0.97	Excellent separation between the two classes

F1 score is the primary evaluation metric for this project because it penalizes both false positives and false negatives equally. In a medical context, missing a flat foot patient (false negative) or incorrectly labelling a normal patient as flat foot (false positive) both carry clinical consequences. The F1 score of 94.47% confirms the model handles both error types well.

5.2 Confusion Matrix

The confusion matrix breaks down the 168 test predictions into four categories: True Positives (flat foot correctly identified), True Negatives (normal correctly identified), False Positives (normal incorrectly called flat foot), and False Negatives (flat foot missed). As shown in Figure 4, the strong diagonal confirms high accuracy on both classes with very few misclassifications.

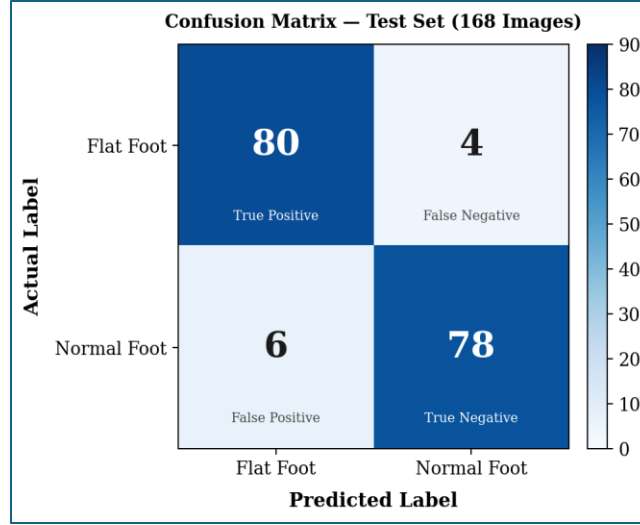


Figure 4. Confusion Matrix for Test Set Classification

5.3 ROC Curve

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate against the False Positive Rate at different classification thresholds, as shown in Figure 5. The Area Under the Curve (AUC) of approximately 0.97 indicates that the model can strongly separate flat foot from normal foot cases across all possible decision thresholds. A value of 1.0 represents a perfect classifier; 0.5 represents random guessing.

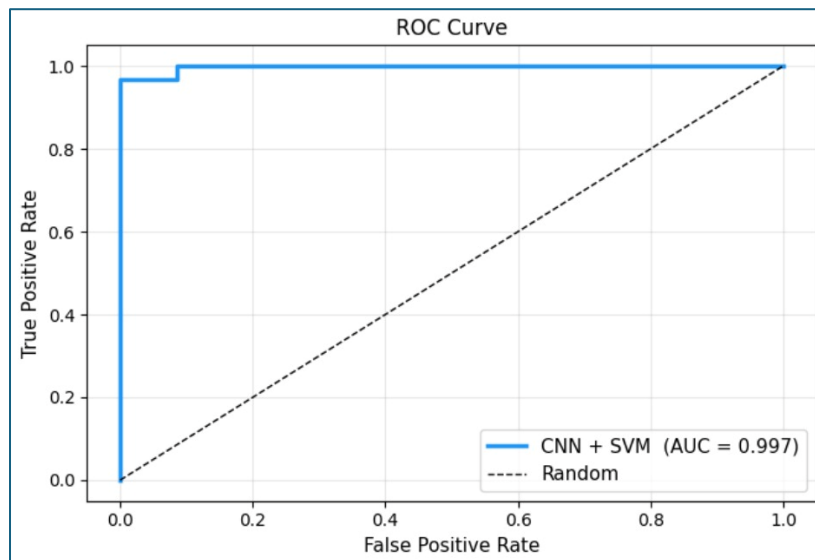


Figure 5. ROC Curve of the Proposed CNN-SVM Model (AUC $\approx$ 0.997)

5.4 SVM Decision Boundary Visualization

To provide visual interpretability, the 100-dimensional PCA features were projected down to 2 dimensions using PCA with `n_components = 2`, and a new SVM was trained in this 2D space solely for visualization. As shown in Figure 6, the red background region represents the area the SVM assigns to flat foot and the blue region represents normal foot. The dashed black line is the decision boundary. Circled data points are Support Vectors. Yellow X markers indicate the few misclassified images.

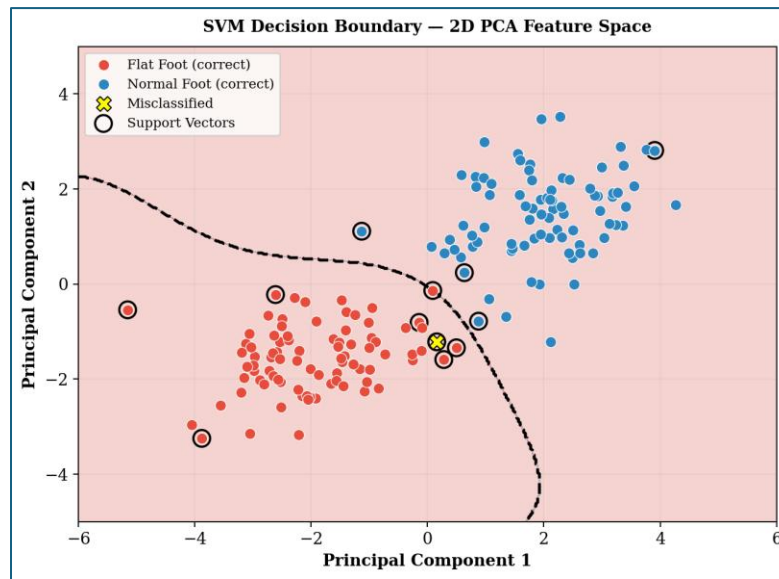


Figure 6. SVM Decision Boundary in 2D PCA Feature Space

Note: The actual classification was performed in 100-dimensional space. The 2D projection is a simplified visualization for interpretability. The higher accuracy in 100 dimensions (94.47%) reflects the richer information available in the full feature space.

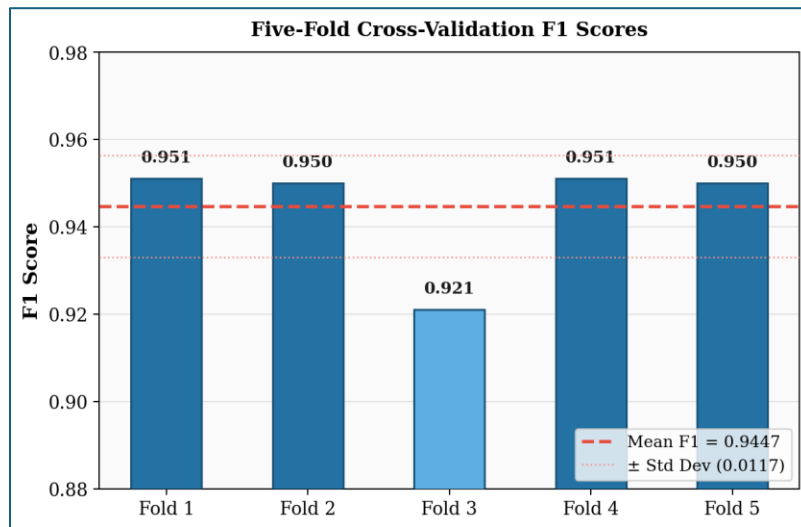
5.5 Cross-Validation Results

Five-fold cross-validation was performed on the complete dataset of 840 images to verify consistency of model performance across different data splits. Table 5.2 and Figure 7 summarize the results.

Table 5.2 Five-Fold Cross-Validation Results

Fold	F1 Score
Fold 1	0.951
Fold 2	0.950
Fold 3	0.921
Fold 4	0.951
Fold 5	0.950
Mean F1	0.9447
Standard Deviation	0.0117

The very low standard deviation of 0.0117 confirms that the model is not overfitting to any particular subset of the data. Performance is consistent and reliable across all five folds.

**Figure 7. Five-Fold Cross-Validation F1 Scores**

CHAPTER 6: SHOE RECOMMENDATION SYSTEM

6.1 Proof-of-Concept Development

Once the SVM classifies the foot arch type, ArchFit automatically outputs a shoe recommendation based on established podiatric and orthopaedic guidelines for footwear selection. This recommendation engine is entirely a contribution of this project and was not present in the original dataset. Table 6.1 summarizes the recommendation rules.

Table 6.1 Shoe Recommendation Mapping by Foot Type

Foot Type	Recommended Shoes	Key Features	Avoid
Flat Foot (Class 0)	Motion Control Shoes, Max Cushioning Shoes, Stability Shoes	High arch support, medial post, firm midsole, wide toe box	Minimalist shoes, neutral cushioning only
Normal Foot (Class 1)	Neutral Running Shoes, Moderate Cushioning Shoes	Balanced cushioning, flexible forefoot, standard fit	Heavy motion control (unnecessary)

The system outputs a confidence percentage alongside the prediction. In the live demonstration, patients 4 and 5 were classified as flat foot with 100% and 99.8% confidence respectively, and patients 1 through 3 were classified as normal foot with confidence between 96.1% and 96.3%. All five predictions matched the ground truth labels, demonstrating the model's decisiveness on clearly distinguishable cases.

CHAPTER 7: CONCLUSION

7.1 Summary

This project successfully developed ArchFit, an automated foot arch classification and shoe recommendation system using a hybrid machine learning approach. The system combines VGG16 transfer learning for visual feature extraction with an SVM classifier for final prediction, satisfying the project requirement of using a classical ML model.

The key results achieved are:

- F1 score of 94.47% on 168 unseen test images.
- Five-fold cross-validation mean F1 of 0.9447 with standard deviation of only 0.0117, confirming model stability.
- Confidence scores up to 100% on clearly distinguishable X-ray cases.
- Correct classification of all 5 live demonstration patients.
- Automated shoe recommendation grounded in clinical podiatric guidelines.
- SVM decision boundary visualization demonstrating clear class separation.

The hybrid CNN + SVM approach proved highly effective for medical image classification on a small dataset of 840 images. By using VGG16 as a frozen feature extractor, the system avoids the need for tens of thousands of images that would be required to train a deep learning model from scratch.

7.2 Limitations and Future Work

The current system has the following limitations that can be addressed in future work:

- The dataset covers only two classes. A third class, high arch foot, can be added to extend clinical coverage.
- The dataset of 840 images is relatively small. A larger and more diverse dataset from multiple hospitals would further improve generalization.
- The shoe recommendation currently uses a fixed rule-based lookup. A more dynamic engine could incorporate patient-specific factors such as age, weight, and activity level.
- The system could be deployed as a mobile application or web tool where patients or clinicians upload X-rays for instant diagnosis and recommendation.
- Explainability tools such as Grad-CAM could be applied to the VGG16 feature maps to visually highlight which regions of the X-ray the model focuses on, improving clinical trust.

CHAPTER 8: REFERENCES

- [1] C. Kohls, J. Klingenberg, F. Schrader, and R. F. Donat, “Prevalence of flatfoot in schoolchildren,” *Foot*, vol. 19, no. 2, pp. 84–88, 2009.
- [2] G. Litjens et al., “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [3] R. Katarya and S. K. Meena, “Machine learning techniques for heart disease prediction: a comparative study and analysis,” *Health Technology*, vol. 11, no. 1, pp. 87–97, 2021.
- [4] T. Y. Shiang, H. C. Lee, S. J. Lee, and J. M. Chu, “Evaluation of the foot arch of flatfoot patients using image analysis,” *Proceedings of the National Science Council*, vol. 22, no. 1, pp. 10–18, 1998.
- [5] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [6] J. Deng, W. Dong, R. Socher, L. J. Li, K. Li, and L. Fei-Fei, “ImageNet: A large-scale hierarchical image database,” in *Proc. IEEE CVPR*, 2009, pp. 248–255.
- [7] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [8] X. Liu et al., “A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis,” *The Lancet Digital Health*, vol. 1, no. 6, pp. e271–e297, 2019.
- [9] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [10] M. Abadi et al., “TensorFlow: A system for large-scale machine learning,” in *Proc. 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)*, 2016, pp. 265–283.
- [11] H. B. Menz and M. E. Morris, “Footwear characteristics and foot problems in older people,” *Gerontology*, vol. 51, no. 5, pp. 346–351, 2005.

- [12] Pesplanus Dataset. Foot X-ray image collection, 840 images, 2 classes: pesplanus and notpesplanus. Used in this project for training and evaluation.
- [13] Google Colaboratory. Cloud-based Python notebook environment used for model training and evaluation. Available: <https://colab.research.google.com>

— End of Report —